

Anker VLM 训练流程 • 阶段输入/输出对齐表

2026-07-09 • 与 docs/WALKTHROUGH.md 衔接图逐字一致 • commit 以 GitHub main 为准

一、准备段(标注线,可与训练完全并行)

#	阶段	输入(来自)	输出	被谁消费
1a	GT 转换 data/euno_wds	euno 标注 json + wds 目录(客户 GCS)	DATA/labels.jsonl	之后所有训练/推理/评测
1b	字节核对 eval/check_format_alignment	3+ 条 GT 原始串(原样复制)	通过/FAIL(闸门,无文件)	放行开训
1b'	机位兜底 data/camera_fingerprint	labels 中 camera=unknown 条目	meta.camera_id 回写	验证集防泄漏切分
1c	监控集 eval/monitor_set	labels.jsonl	monitor.jsonl	每阶段结束的离线验收闸门
2A	Gemini 标注 annotation/label_euno_wds	wds 帧 +(可选)标注 json 限定范围	pass1.jsonl	2A' / 2A''
2A'	gt 过滤 annotation/consistency_filter	pass1 + labels(GT)	filtered/whitelist_ids.txt	2A''
2A''	资产拆分 annotation/split_assets	pass1 + whitelist	DATA/asset_A_attributes.jsonl DATA/asset_C_reasoning.jsonl DATA/asset_D_whitelist.txt	5(A)/ 6(C);D 仅留档 样本永远全量,白名单不裁样本

二、训练段(100k balanced 轮)

#	阶段	输入(来自)	输出(final/ 内容)	被谁消费
3a	Projector warmup	labels	phase5_sft_a/final (仅 projector.pt ,无 adapter)	3b --init-from
3b	SFT 联合	labels + 3a final	phase5_sft_b/final (adapter 两件套 + projector.pt)	4a --ckpt;4b --init-from
4a	v0 推理	3b final + labels	preds_v0.jsonl	4b
4b	Hard Mining 续训	preds_v0 + labels → sw.jsonl;+3b final	phase5_sft_hm/final (必须 --output 换目录!)	5 --init-from
5	辅助头(5b)	labels 全量 + asset_A + hm final	phase5b_aux/final (+aux_heads.pt,不交付)	6(yaml init_from)
6	隐式 CoT(5d)	labels 全量 + asset_C + 5 final	phase5d_cot/final (= v1.5)	4a' ckpt;7b init; 8 的 --projector 来源
4a'	v1.5 再推理	6 final + labels	preds_v15.jsonl (不能复用 preds_v0!)	7a + think 泄漏验收
7a	KTO 数据	preds_v15 + labels(客户数据不传 whitelist)	outputs/kto_data.jsonl	7b

7b	KTO	kto_data + 6 final	phase6_kto/{checkpoint-*, final}	7c
7c	SWA	7b 的 checkpoint-*	swa_final (adapter 平均 + projector 自动附带)= v2	8
8	导出	swa_final + base 权重 + 6 final/projector.pt	deliverables/llm_adapter/ (→rkllm) vision_merged.pt (→onnx→rknn)	客户端侧编译部署

三、1M 自然分布轮(第二轮,真续训)

#	阶段	输入(来自)	输出	被谁消费
9	1M 续训 configs/phase7_1m_calib	1M GT 标注(客户 D7 交付)转换的 labels_1m + swa_final(v2);LR×0.5,≤2ep 早停	phase7_1m_calib/final	1M 上重复 4a/4b(HM)
9'	1M HM + KTO + SWA	同 100k 轮衔接方式,数据换 1M	新 swa_final (= 最终模型)	Step 8 重新导出 = 最终交付

⚠ 三个最易接错的衔接点(审计中真实出过错): ① 7a 必须用 preds_v15(v1.5 的错例),不能复用 preds_v0; ② hard mining 续训必须 --output 换目录,否则覆盖 3b 产物; ③ Step 8 的 --projector 必传(6 final/projector.pt),否则视觉端丢训练成果。

通用规则:所有 --ckpt/--init-from 都指上一步的 final/;任何阶段监控集验收不达标 → 弃该产物,下一阶段直接 init 上一个达标 final(回退协议)。

四、各阶段数据文件格式与示例(逐字段)

4.1 euno 原始标注 json —— 客户输入(LlamaFactory 对话格式,1a 消费)

```
[
  {
    "id": "8000",
    "videos": ["trainset_4k_250828/5911b3bd-ae4d-4d1a-9aaf-91a1462a5ae6_segment_1"],
    "conversations": [
      {
        "from": "human",
        "value": "<video>You are an AI surveillance analyst. ...(生产 prompt 全文,所有样本相同)",
      },
      {
        "from": "gpt",
        "value": "C\n|During the day, a man in light-colored clothes took a package from a residential doorway and left."
      }
    ]
  }
]
```

要点: videos[0] = index.json 的 key(定位 wds 分片); gpt.value = GT 串,分隔符为 **无空格的 |**(已 5/5 字节核对)。

4.2 DATA/labels.jsonl —— 通用主标签(1a 产出;3a/3b/4b/5/6/7b 全消费)

```
{
  "video_id": "trainset_4k_250828/5911b3bd-ae4d-4d1a-9aaf-91a1462a5ae6_segment_1",
  "video_uri": "trainset_4k_250828/5911b3bd-ae4d-4d1a-9aaf-91a1462a5ae6_segment_1",
  "duration_sec": null,
  "resolution": "384x384",
  "labels": {
    "role_type": "C",
    "sub_keyscene": "n",
    "description": "During the day, a man in light-colored clothes took a package from a residential doorway and left."
  },
  "meta": {
    "camera_id": "unknown",
    "storage": "wds",
    "wds_dir": "/data/anker_video_clips_wds_full",
    "shard": 15,
    "euno_id": "8000"
  }
}
```

要点: ① meta.storage="wds" → 训练/推理自动走分片帧读取,无需 mp4;② camera_id 从文件名设备序列号(T8xxx)提取,uuid 命名 → "unknown" → 跑 camera_fingerprint 兜底(切分时 unknown 按单条处理);③ duration/分辨率被上游预处理抹掉,固定 384x384;④ 代理集(视频文件)变体: meta 无 storage 字段,video_uri 为文件名,去 data.video_root 下解码。

4.3 gt_samples.txt —— 字节核对输入(1b;从数据文件原样复制,禁手敲)

```
D|g|A man in a hat approached a camera at a residence, paused, then turned and left.
B|i|Daytime at a residence, a male worker puts down a package, pauses, then picks it up and continues delivering.
E|p|At night, outside the residence, a bear knocked over a blue bucket placed on the porch.
```

4.4 pass1.jsonl —— Gemini 增强标注产物(2A;真实实测记录)

```
{
  "video_id": "trainset_ucsd/decrypt_T8030TEST0001_ped2_train001_segment_1",
  "gemini_model": "gemini-2.5-flash",
  "temperature": 0.1,
  "gemini_output": {
    "attributes": {
      "clothing": "casual",
      "held_items": "none",
      "posture": "walking",
      "action": "walking",
      "view_type": "non_doorbell",
      "sub_role_type": "Passerby",
      "dwell": "passes_through"
    },
    "reasoning_chain": "[身份线索] 多名行人穿着休闲,无特定身份标识 [场景线索] 户外步道,行人正常通行 [结论] 未指定身份的其他正常活动 → D|m",
    "predictions": {
      "role_type": "D",
      "sub_keyscene": "m",
      "description": "Multiple pedestrians walk across a paved outdoor area from an overhead camera view."
    }
  }
}
```

要点: ① attributes 七键的取值 = data/taxonomy.AUX_VOCABS 枚举(训练两端单一来源);② predictions 只用于 2A' 一致性过滤,不作训练标签(GT 全量信任);③ 失败记录写同名 .errors 文件(含 errors 字段),重跑自动重试。

4.5 filtered/whitelist_ids.txt(2A')与 4.6 资产文件(2A')

```
# whitelist_ids.txt — 每行一个 video_id;语义 = "该样本的 Gemini 增强标注可信"(非 GT 对错证据)
trainset_4k_250825/T8030P2322510409_T8160P2122501B2C_20230719_segment_1

# asset_A_attributes.jsonl — Phase 5b 辅助头监督(仅白名单内条目)
{"video_id": "trainset_4k_250825/T8030P2322510409_..._segment_1", "confidence": 1.0,
 "attributes": {"clothing": "uniform", "held_items": "package", "posture": "walking",
               "action": "placing_item", "view_type": "doorbell",
               "sub_role_type": "DeliveryPerson", "dwell": "brief_stop"}}

# asset_C_reasoning.jsonl — Phase 5d 隐式 CoT 目标(仅白名单内条目)
{"video_id": "trainset_4k_250825/T8030P2322510409_..._segment_1",
 "reasoning_chain": "[身份线索] 制服+径直走向门口 [场景线索] 手持包裹放置于门垫 [结论] 工作人员投递包裹 → B|i"}

```

要点: 资产层已按白名单过滤 —— 训练样本永远全量;无资产/低置信(<0.5)样本自动降级: aux 标签 -100(该头 loss=0)、CoT 走生产模式,样本不丢。

4.7 训练目标串 —— 模型实际学习的 label 段(3b/4b/5 用 ①;6 用 ①+②混合)

```
① 生产模式(全阶段):
B|i|A uniformed delivery person places a package at the front door and walks away.
  ^^^ RT字母+两个|+SK字母 → loss 权重 ×4;描述段 ×1;前缀 prompt 段全部 -100

② 推理链模式(仅 5d,占 60%,prompt 尾加 [REASON] 标记):
<think>[身份线索] 制服+径直走向门口 [场景线索] 手持包裹放置 [结论] B|i</think>
B|i|A uniformed courier places a parcel at the door and leaves.
  <think> 段 loss 权重 0(只作条件不反传);最后 0.5 epoch 强制全 ①(退火防泄漏)

```

4.8 preds_v0 / preds_v15.jsonl(4a / 4a' 推理所产物)与 sw.json(4b)

```
# preds *.jsonl — 贪心解码原始输出;断点续跑按 video_id 去重追加
{"video_id": "trainset_.../..._segment_1", "output": "C|s|A person lingers near the front door at night."}

# sw.json — hard mining 样本权重(video_id → 权重)
{"trainset_.../seg_1": 1.0, "trainset_.../seg_7": 3.0, "trainset_.../seg_9": 1.3}

```

要点: 1.0=答对样本;3.0=可信错例;分数权重(如 1.3)= per-class 膨胀封顶(1.5×)压缩后的值,物理复制用流式最大余数法保证总质量守恒。

4.9 kto_data.jsonl(7a → 7b)

```
{"video_id": "trainset_.../seg_1", "completion": "B|i|A courier leaves a parcel at the front door.", "label": true}
{"video_id": "trainset_.../seg_7", "completion": "D|m|A person walks past the porch casually.", "label": false}

```

要点: label=true → desirable = GT 完整输出串(含 GT 描述,隐式保住描述质量); false → undesirable = v1.5 分类判错的原始输出(RT 或 SubKS 任一字母错即入选,纯规则判定零标注成本)。训练时每 batch 由分层采样保证混入错例。

4.10 euno 推理结果 json —— 基线复现输入(eval/euno_results_adapter 消费)

```
[{"id": "3", "video": "testset_1k_250730/00e22ff1-..._segment_1.mp4",
 "conversations": [{"from": "human", "value": "...(prompt)"},
                   {"from": "gpt", "value": "B|i|Daytime at a residence, a male worker puts down a package, ..."}],

```

```
"pred": {"result": "B|i|A person in a blue vest approaches a walkway camera, ...",  
        "score": [0.9472, 1.0]}}
```

要点: adapter 拆成 preds.jsonl + gt.jsonl 喂 eval/metrics,即客户基线 93.60/76.14/86.22 的同口径复现路径。